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Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

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Federated Heterogeneous Graph Neural Networks for TBML Detection

A Major Project Report (CS481P)

Submitted by,

Nikhil Vasu	**1RV22CS128**
Rohith Biradar	**1RV22CS164**
Suraj Chanaveeragoudra	**1RV22CS211**

Under the guidance of

Dr. Nagaraj G. S
Professor
Dept. of CSE
RV College of Engineering

Prof. Sanjana Ravindra Otihal
Assistant Professor
Dept. of CSE
RV College of Engineering

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Bachelor of Engineering in

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(Autonomous institution affiliated to VTU, Belagavi)

Department of Computer Science and Engineering

CERTIFICATE

Certified that the major project (CS481P) work titled ***Federated Heterogeneous Graph Neural Networks for TBML Detection*** is carried out by **Nikhil Vasu** (**1RV22CS128**), **Rohith Biradar** (**1RV22CS164**) and **Suraj Chanaveeragoudra** (**1RV22CS211**) who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in **Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

Guide	Head of the Department	Principal
Dr. Nagaraj G. S	Dr. Shanta Rangaswamy	Dr. K. N. Subramanya

External Viva

Name of Examiners	Signature with Date
1.	
2.	

DECLARATION

We, **Nikhil Vasu**, **Rohith Biradar** and **Suraj Chanaveeragoudra** students of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Federated Heterogeneous Graph Neural Networks for TBML Detection**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in **Computer Science and Engineering** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

	Name	Signature
1.	Nikhil Vasu(1RV22CS128)	
2.	Rohith Biradar(1RV22CS164)	
3.	Suraj Chanaveeragoudra(1RV22CS211)	

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ABSTRACT

Trade-Based Money Laundering (TBML) has become a major challenge in modern financial systems due to the increasing complexity and scale of global trade transactions. Conventional anti-money laundering systems primarily rely on rule-based monitoring and traditional machine learning techniques, which are often inadequate for identifying hidden transactional dependencies and evolving laundering strategies. These approaches suffer from limitations such as high false positive rates, poor relational learning capability, and reduced scalability in large financial networks. To address these issues, this report presents a Federated Heterogeneous Graph Neural Network (HGNN) framework for real-time TBML detection capable of modeling complex trade relationships while preserving institutional data privacy.

The primary objective of this work is to develop a scalable and privacy-preserving TBML detection architecture using graph-based deep learning methodologies. The proposed framework represents financial ecosystems as heterogeneous graphs consisting of entities such as accounts, importers, exporters, transactions, and trade documents. Graph Attention Network (GAT)-based convolutional layers are employed to learn structural and semantic relationships between interconnected entities for effective anomaly detection. Federated learning mechanisms are incorporated to facilitate collaborative model training across distributed financial institutions without centralized data sharing. The computational framework includes graph construction, feature extraction, node embedding generation, suspicious activity classification, and real-time risk score prediction.

The proposed system is implemented using Python, PyTorch Geometric, FastAPI, Memgraph, PostgreSQL, and React.js. Experimental evaluation is carried out on simulated TBML datasets containing both legitimate and suspicious trade transactions. Performance analysis is conducted using standard classification metrics including accuracy, precision, recall, and F1-score. The experimental results indicate that the proposed framework effectively captures hidden laundering patterns and improves fraud detection capability when compared with conventional machine learning approaches. The architecture further supports scalable deployment, reduced false positive rates, and efficient real-time monitoring of high-volume trade transactions.

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Introduction to Trade-Based Money Laundering

CHAPTER 1

INTRODUCTION TO TRADE-BASED MONEY LAUNDERING

The rapid growth of international trade and digital financial infrastructure has significantly increased the complexity of monitoring global financial transactions. Alongside economic globalization, financial institutions are facing increasingly sophisticated forms of illicit financial activities that exploit weaknesses in conventional Anti-Money Laundering (AML) systems. Among these, Trade-Based Money Laundering (TBML) has emerged as one of the most challenging financial crimes due to its ability to conceal illegal fund transfers within legitimate trade operations. The distributed and interconnected nature of modern trade ecosystems necessitates intelligent, scalable, and privacy-preserving analytical frameworks capable of identifying hidden transactional relationships in real time.

> *"For financial institutions, TBML is uniquely challenging because illicit funds are embedded within legitimate trade transactions, leaving traditional AML controls blind to the risks."*
>
> *– Dr. Sebastian Hetzler, IMTF Co-CEO [1]*

As highlighted in the above statement, Trade-Based Money Laundering exploits the complexity of international commerce by embedding illicit financial flows within apparently legitimate trade transactions. This significantly limits the effectiveness of conventional compliance systems that primarily rely on isolated transaction monitoring and static rule-based verification mechanisms.

1.1 Overview of Trade-Based Money Laundering

Trade-Based Money Laundering (TBML) is defined as the process of disguising the proceeds of crime and transferring value through legitimate international trade transactions in an attempt to legitimize illicit financial activities [2]. Unlike conventional money laundering mechanisms that rely on direct movement of funds, TBML conceals illegal financial flows within normal commercial trade operations such as import-export activities, invoice settlements, and cross-border transactions. Due to the involvement of multiple intermediaries, financial institutions, customs agencies, and international jurisdictions, identifying suspicious trade behavior becomes significantly difficult for conventional Anti-Money Laundering systems.

1.1.1 Global Financial Crime Scenario

The increasing volume of international trade and cross-border financial transactions has considerably complicated the process of monitoring illicit financial activities. According to the United Nations Office on Drugs and Crime (UNODC), money laundering activities account for nearly 2% to 5% of the global Gross Domestic Product (GDP), corresponding to approximately $800 billion to $2 trillion annually [3]. Despite the deployment of large-scale compliance frameworks and transaction monitoring systems, Europol reports that global authorities successfully intercept less than 1% of illicit financial flows [4]. These statistics indicate the growing sophistication of modern laundering networks and the limitations of existing monitoring infrastructures.

The large-scale movement of goods, services, and financial assets across international markets creates an environment where illicit transactions can be concealed within legitimate commercial activities. As financial institutions continue to strengthen Know-Your-Customer (KYC) regulations and banking surveillance systems, criminal organizations increasingly exploit trade ecosystems to bypass conventional financial monitoring mechanisms.

Table 1.1: Key Statistics Related to Global Trade-Based Financial Crime

Financial Crime Metric	**Estimated Value**
Global GDP associated with money laundering annually	2% – 5%
Estimated annual TBML economic volume	$1.6T – $2.0T
Illicit flows linked to trade misinvoicing in developing economies	87%
Global interception rate of illicit financial flows	<1%
Reduction in false positive alerts using AI-driven monitoring systems	Up to 40%

1.1.2 Trade-Based Money Laundering

Trade-Based Money Laundering enables criminal organizations to transfer illicit value through apparently legitimate trade transactions while avoiding direct movement of cash. Common TBML techniques include over-invoicing, under-invoicing, multiple invoicing, false description of goods, phantom shipments, and the use of shell entities to manipulate commercial trade records. Since these activities are embedded within genuine trade operations involving multiple organizations and jurisdictions, detecting suspicious transactional behavior becomes highly complex for financial institutions and regulatory agencies.

Reports published by Global Financial Integrity indicate that trade misinvoicing and related TBML activities account for nearly 87% of illicit financial flows originating from developing economies [5]. Similarly, industry analyses estimate that approximately $1.6 trillion to $2 trillion in illicit value is concealed annually within trade invoices and commercial documentation [1]. These findings demonstrate the growing dependence of organized criminal networks on international trade systems for laundering illicit financial assets.

1.1.3 Problem Area

Conventional Anti-Money Laundering systems primarily rely on rule-based monitoring frameworks, predefined transaction thresholds, and manual auditing procedures for identifying suspicious financial activities. Although such approaches are effective in detecting isolated anomalies, they often fail to identify hidden relationships among entities involved in coordinated laundering operations. Modern TBML networks operate through interconnected ecosystems involving importers, exporters, shell companies, offshore entities, logistics providers, and financial intermediaries distributed across multiple jurisdictions, making detection significantly more challenging.

Traditional machine learning approaches further suffer from limitations associated with isolated transaction analysis and lack of relational understanding. Since financial transactions are processed independently, existing systems are unable to effectively capture structural interactions and behavioral dependencies hidden within large-scale trade networks. As a result, conventional AML systems frequently generate high false positive rates, increased compliance overhead, and delayed investigation processes.

Another major challenge involves institutional privacy and regulatory compliance constraints. Financial organizations are often unable to share sensitive transactional data across institutions due to strict legal and regulatory restrictions. Consequently, centralized fraud detection architectures become impractical in real-world banking environments. These limitations highlight the need for intelligent, scalable, and privacy-preserving analytical frameworks capable of effectively identifying suspicious transactional behavior across distributed financial ecosystems.

1.2 Literature Review

The increasing complexity of global financial systems and the rapid growth of international trade have motivated extensive research in Anti-Money Laundering (AML) and Trade-Based Money Laundering (TBML) detection systems. Conventional AML frameworks primarily relied on rule-based transaction monitoring and manual auditing procedures to identify suspicious financial activities. However, the emergence of interconnected financial ecosystems, cross-border trade mechanisms, and sophisticated laundering strategies has significantly reduced the effectiveness of traditional approaches. Consequently, recent research has increasingly focused on integrating machine learning, graph analytics, and distributed intelligence techniques into modern AML systems.

Jiani Fan, Lwin Khin Shar, Ruichen Zhang, Ziyao Liu, Wenzhuo Yang, Dusit Niyato, and Kwok-Yan Lam [6] presented a comprehensive survey on deep learning-based Anti-Money Laundering frameworks for mobile financial systems. The study explored the application of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Graph Neural Networks (GNNs) for transaction anomaly detection and proposed a Context-Risk-Predict framework for improving AML intelligence. The authors demonstrated that deep learning architectures significantly improve predictive performance compared to conventional rule-based systems. However, the framework primarily focused on centralized transaction processing environments and did not address institutional privacy constraints or distributed financial data challenges.

Morten Hjelholt Jensen and Alexandros Iosifidis [7] investigated the use of statistical analysis and supervised machine learning algorithms for financial anomaly detection. Their study evaluated models such as Logistic Regression, Random Forests, and Gradient Boosting techniques for suspicious transaction classification. Experimental results showed improved detection capability compared to static rule-based approaches. Nevertheless, the proposed methods analyzed transactions independently and lacked the ability to capture relational dependencies among interconnected financial entities involved in coordinated laundering operations.

Jinlong Song, Zhen Guo, Yutian Lei, and Minghao Zhao [8] proposed a dimension expansion framework for learning hidden money laundering activities within transaction networks. The framework enhanced transaction representations by integrating neighboring account information and contextual interaction features. The study demonstrated that relational feature aggregation improves the detection of concealed laundering patterns across financial networks. However, the model depended heavily on manually engineered relational features and lacked scalability for distributed cross-institutional financial environments.

In another study, Jinlong Song, Haoran Chen, Peng Xu, and Minghao Zhao [9] introduced a Social Attention Graph Neural Network framework for detecting illicit accounts in blockchain transaction systems. The proposed architecture transformed transaction ecosystems into graph structures and utilized attention-based graph learning mechanisms for anomaly classification. The study demonstrated that Graph Neural Networks outperform conventional machine learning models in identifying suspicious transactional behavior. However, the framework focused primarily on cryptocurrency transaction ecosystems and did not address invoice-level trade manipulations or heterogeneous commercial trade relationships associated with TBML activities.

Xiang Li, Zhen Wang, Yong Zhou, and Lei Zhang [10] proposed a Multi-View Graph-Based Hierarchical Representation Learning framework for organized money laundering group detection. Their work utilized multiple graph perspectives and hierarchical embedding mechanisms to identify coordinated laundering communities within financial transaction networks. Experimental findings demonstrated improved graph representation capability for identifying hidden laundering structures. However, the framework relied on predefined graph meta-path relationships and centralized graph repositories, limiting its applicability in privacy-sensitive financial environments.

Eren Gezici and Emre Sefer [11] proposed AMLPD, an unsupervised Anti-Money Laundering detection framework utilizing Personalized PageRank and graph diffusion techniques for deep vertex representation learning. The framework effectively captured hidden graph structures without requiring labeled datasets and demonstrated strong anomaly detection capability in sparse financial networks. However, the absence of supervised optimization mechanisms resulted in increased false positive classifications within highly imbalanced financial transaction datasets.

Mohammad Karim, Tariq Hassan, and Sameer Ali [12] investigated scalable semi-supervised graph learning architectures for large-scale financial transaction monitoring. Their study evaluated models such as SkipGCN and EvolveGCN for node classification within AML transaction graphs. Experimental results demonstrated efficient scalability and improved anomaly detection performance in large graph repositories. Nevertheless, the framework assumed centralized access to financial transaction graphs and did not address institutional privacy and regulatory constraints associated with collaborative fraud detection systems.

Jongmin Koo, Hyun Park, and Seungwoo Kim [13] proposed an autoencoder-based suspicious transaction detection framework using Risk-Based Approaches (RBA) for banking systems. The proposed model reduced anomaly detection complexity by learning latent transactional representations from financial datasets. Although the framework improved internal banking surveillance efficiency, it focused primarily on point anomalies associated with individual accounts and failed to model interconnected laundering relationships distributed across complex trade ecosystems.

The literature survey indicates a clear transition from conventional rule-based AML systems toward machine learning, graph neural networks, and intelligent graph analytics for financial crime detection. Existing studies demonstrate that graph-based architectures significantly improve the ability to identify hidden transactional relationships and coordinated laundering structures compared to traditional machine learning techniques. However, several limitations remain insufficiently addressed, including heterogeneous graph modeling, real-time transaction analysis, institutional privacy preservation, and scalability across distributed financial environments. Most existing approaches either rely on centralized transaction repositories or focus primarily on banking or cryptocurrency transaction datasets without effectively modeling the complex trade relationships associated with Trade-Based Money Laundering activities.

Furthermore, many existing AML frameworks process financial transactions in isolation and fail to capture the interconnected dependencies among importers, exporters, shell entities, logistics providers, and trade documents distributed across multiple jurisdictions. The absence of privacy-preserving collaborative intelligence mechanisms further limits the practical deployment of such systems in real-world financial ecosystems where direct sharing of sensitive transactional data is restricted by regulatory policies. In addition, limited focus has been given to integrating heterogeneous graph learning, federated intelligence, and real-time anomaly detection within a unified Anti-Money Laundering framework. These research gaps highlight the need for an intelligent, scalable, and privacy-preserving architecture capable of modeling complex trade ecosystems while supporting collaborative real-time TBML detection across distributed financial institutions.

1.3 Motivation

The work, *"Federated Heterogeneous Graph Neural Networks for Real-Time Detection of Trade-Based Money Laundering"*, was chosen to address the increasing complexity of financial crimes within modern global trade ecosystems through the development of an intelligent and privacy-preserving fraud detection framework. Conventional Anti-Money Laundering (AML) systems primarily rely on rule-based monitoring and isolated transaction analysis methods that are often ineffective in identifying hidden relationships and coordinated laundering activities distributed across interconnected financial networks. Although such systems are widely deployed in banking environments, they generate high false positive rates, require extensive manual investigation, and face significant challenges in detecting sophisticated Trade-Based Money Laundering (TBML) activities in real time.

The proposed work aims to overcome these limitations by developing a Federated Heterogeneous Graph Neural Network framework capable of modeling complex transactional relationships among financial entities while preserving institutional privacy. The integration of heterogeneous graph learning, graph attention mechanisms, and federated intelligence enables collaborative fraud analysis without direct sharing of sensitive transactional data across organizations. In addition, the framework supports automated Suspicious Transaction Report (STR) generation and dashboard-based visualization for real-time fraud monitoring and compliance analysis, demonstrating the feasibility of scalable and AI-driven Anti-Money Laundering systems for modern financial environments.

1.4 Problem Statement

Existing Anti-Money Laundering (AML) systems primarily rely on rule-based monitoring and isolated transactional analysis, making them ineffective in detecting complex Trade-Based Money Laundering (TBML) activities and hidden financial relationships. In addition, privacy and regulatory constraints restrict collaborative fraud detection across financial institutions. Therefore, there is a need for a scalable and privacy-preserving framework capable of modeling interconnected financial data and accurately identifying suspicious transactional behavior in real time.

1.5 Objectives

The objectives of the project are

1. To develop a Federated Heterogeneous Graph Neural Network framework for real-time detection of Trade-Based Money Laundering activities.

2. To model complex transactional relationships among financial entities using heterogeneous graph representation and graph attention mechanisms.

3. To generate Suspicious Transaction Reports (STRs) for identified fraudulent transactions to support financial investigation and compliance processes.

4. To design an interactive dashboard for visualization of fraud predictions, transaction analysis, and real-time monitoring of suspicious activities.

1.6 Brief Methodology of the project

The proposed methodology focuses on developing a privacy-preserving and graph-based framework for real-time Trade-Based Money Laundering (TBML) detection. Initially, financial transaction data, trade records, customer information, and entity relationships are collected and preprocessed to remove inconsistencies and prepare structured datasets for analysis. The processed data is then transformed into a heterogeneous graph structure where entities such as accounts, organizations, invoices, and transactions are represented as interconnected nodes and edges.

Graph Attention-based learning mechanisms are employed to capture hidden relational dependencies and suspicious transactional patterns within the financial ecosystem. To preserve institutional privacy, federated learning is integrated into the framework, enabling collaborative model training across distributed financial environments without direct sharing of sensitive transactional data. The detected suspicious activities are further utilized for automated Suspicious Transaction Report (STR) generation and visualization through an interactive dashboard for real-time monitoring and fraud analysis.

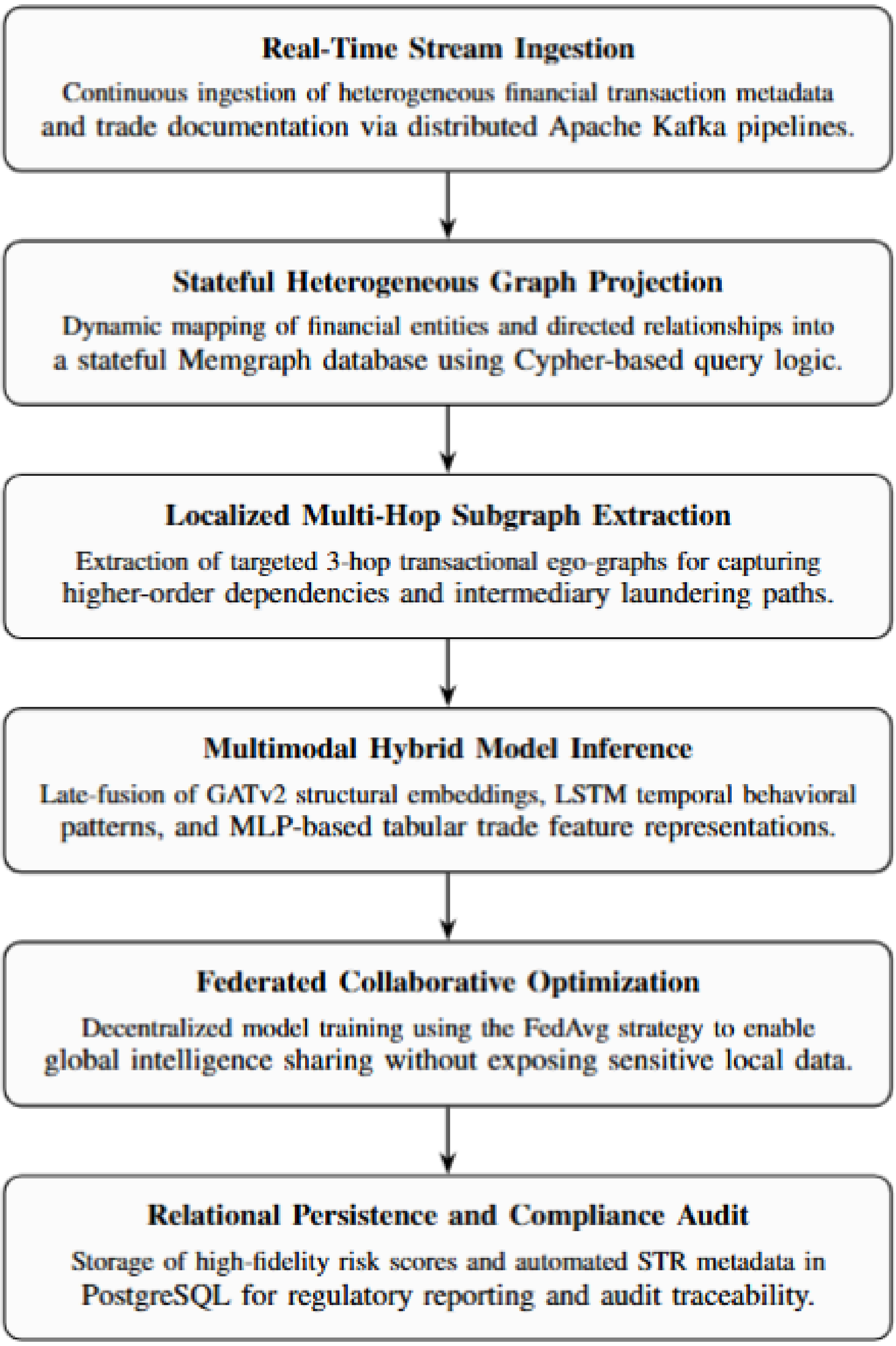

Figure 1.1: Proposed Methodology for TBML Detection

1.7 Assumptions and Constraints of the Work

Assumptions

- The financial transaction records, trade invoices, and customer-related information used for analysis are assumed to be sufficiently structured, consistent, and preprocessed for effective model training and evaluation.

- It is assumed that participating financial institutions maintain similar transactional formats and standardized trade-related attributes required for graph construction and suspicious activity analysis.

- The generated financial datasets are assumed to contain sufficient normal and suspicious transactional patterns necessary for learning hidden behavioral relationships associated with Trade-Based Money Laundering activities.

- It is assumed that heterogeneous graph representations can effectively model relationships among accounts, organizations, transactions, invoices, and trade entities for identifying suspicious financial behavior.

Constraints

- Limited access to real-world Trade-Based Money Laundering datasets due to institutional privacy regulations and confidentiality policies may affect the diversity of transactional data used for evaluation.

- Large-scale graph construction and real-time transaction analysis may increase computational complexity and memory requirements during fraud detection operations.

- The performance of the proposed framework depends significantly on the quality, completeness, and accuracy of the financial data available for model training and testing.

- The proposed framework is validated primarily under controlled academic and experimental conditions and may require additional optimization for large-scale industrial deployment.

1.8 Organization of the Report

This report is organized as follows.

- Chapter 2 discusses the theoretical concepts related to Anti-Money Laundering systems, Trade-Based Money Laundering, Graph Neural Networks, heterogeneous graph learning, and federated learning techniques used in the proposed work.
- Chapter 3 discusses the Software Requirement Specification (SRS) of the proposed system, including functional requirements, performance requirements, software and hardware requirements, and system constraints.
- Chapter 4 discusses the high-level architecture and overall system design of the proposed framework, including workflow diagrams, system components, dashboard architecture, and data flow representation.
- Chapter 5 discusses the detailed design of the proposed system, including graph construction methodology, fraud detection modules, STR generation workflow, and dashboard integration.
- Chapter 6 discusses the implementation details of the proposed framework, including dataset preprocessing, graph generation, backend integration, model development, and dashboard implementation.
- Chapter 7 discusses the testing and validation procedures performed on the proposed system, including functional testing, fraud prediction analysis, dashboard testing, and model evaluation techniques.
- Chapter 8 discusses the experimental results and performance evaluation of the proposed framework using various fraud detection metrics and comparative analysis.
- Chapter 9 discusses the conclusion of the proposed work, existing limitations, and possible future enhancements for improving the framework.

Chapter 2

Theory and Fundamentals of Project

CHAPTER 2

THEORY AND FUNDAMENTALS OF PROJECT

Developing an intelligent framework for Trade-Based Money Laundering detection requires a strong foundation in financial crime analysis, graph theory, and machine learning techniques. This chapter presents the fundamental concepts associated with Trade-Based Money Laundering, Graph Neural Networks, Graph Attention mechanisms, heterogeneous graph representation, and federated learning. The chapter further discusses the computational principles involved in relational transaction modeling, privacy-preserving learning, and real-time fraud analytics. These concepts collectively form the theoretical foundation for the proposed fraud detection framework.

2.1 Trade-Based Money Laundering

Trade-Based Money Laundering (TBML) is a sophisticated form of financial crime in which illicit funds are transferred and concealed through seemingly legitimate international trade activities. Unlike conventional laundering methods that primarily depend on direct financial transactions, TBML exploits import-export systems, trade invoices, shipping documents, and cross-border commercial operations to disguise the movement of illegal funds. Due to the increasing implementation of strict Anti-Money Laundering (AML) regulations and Know Your Customer (KYC) policies, criminal organizations have increasingly shifted toward trade-oriented laundering mechanisms that are comparatively difficult to detect using traditional monitoring systems.

The complexity of modern global trade ecosystems, combined with the involvement of multiple financial entities and transactional relationships, has made TBML detection a major challenge for financial institutions and regulatory agencies. Consequently, intelligent analytical techniques capable of identifying hidden transactional dependencies and suspicious financial behavior have become increasingly important in modern Anti-Money Laundering systems.

2.1.1 Introduction to TBML

Trade-Based Money Laundering primarily involves the manipulation of international trade transactions to illegally transfer value between entities located across different countries. Criminal organizations often misuse trade documentation, invoice values, shipment quantities, and commercial agreements to move illicit funds while making the transactions appear legitimate within global financial systems. Since these laundering activities are embedded within regular import-export operations, identifying suspicious behavior becomes highly challenging for financial institutions and regulatory agencies.

In many TBML operations, colluding buyers and sellers intentionally manipulate the declared value of traded goods to transfer illegal financial assets across borders. Techniques such as over invoicing, under invoicing, phantom shipments, and multiple invoicing are commonly used to conceal the movement of illicit proceeds. Such laundering mechanisms exploit the complexity and high transactional volume of international trade systems, making traditional rule-based Anti-Money Laundering approaches comparatively ineffective in detecting hidden financial relationships.

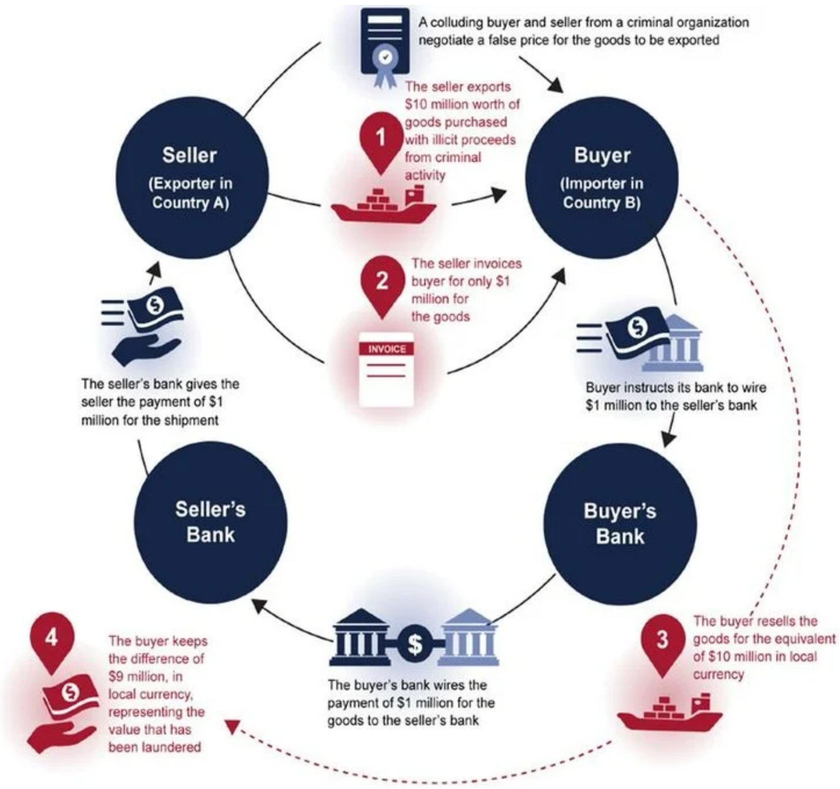

Figure 2.1: Trade-Based Money Laundering through trade misinvoicing and open-account transactions

Source: Adapted from "The Role of Misinvoicing in the Money Laundering Cycle," ResearchGate.

Figure ?? illustrates a typical Trade-Based Money Laundering workflow involving collusion between exporters and importers across different jurisdictions.

2.1.2 Common Techniques Used in TBML

Trade-Based Money Laundering activities are generally performed through the manipulation of trade documentation, shipment details, pricing structures, and financial transactions associated with international trade operations. Criminal organizations exploit the complexity of global trade ecosystems to transfer illicit value across borders while making the transactions appear legitimate. Several laundering techniques are commonly used within modern trade systems to disguise the movement of illegal funds and avoid detection from financial monitoring authorities.

Over Invoicing

Over invoicing is a laundering technique in which the value of exported or imported goods is intentionally declared higher than the actual market value. In this process, the buyer transfers an inflated payment amount to the seller, allowing excess funds to be moved across borders under the appearance of legitimate trade transactions. Criminal organizations use this technique to transfer illicit financial assets while avoiding suspicion from conventional banking systems.

Under Invoicing

Under invoicing involves declaring a lower value for traded goods when compared to their actual market price. This technique allows the buyer or seller to secretly transfer additional financial value outside formal banking channels. Under invoicing is commonly used to evade taxes, conceal illegal profits, and facilitate hidden movement of illicit assets across jurisdictions.

Multiple Invoicing

Multiple invoicing refers to the repeated use of the same trade invoice to justify several financial transactions for a single shipment of goods. By duplicating invoices across multiple banking channels or financial institutions, criminal entities can move large amounts of illicit funds while creating the appearance of legitimate trade payments. Detecting such duplicated transactional behavior is challenging in traditional rule-based monitoring systems.

Phantom Shipments

Phantom shipment laundering occurs when financial transactions are conducted for goods that are never actually shipped or delivered. Fraudulent trade documentation and fabricated shipping records are used to create the illusion of legitimate trade activity while transferring illicit funds between colluding entities. Since no physical goods are exchanged, identifying phantom shipment activities requires strong transactional verification and relational analysis mechanisms.

Misrepresentation of Goods

Misrepresentation of goods involves intentionally providing false descriptions regarding the type, quantity, quality, or category of traded products within commercial documents and invoices. Criminal organizations use this technique to manipulate trade valuations and conceal suspicious financial activities under legitimate business transactions. Such inconsistencies are difficult to identify using isolated transactional analysis methods due to the large volume and diversity of international trade operations.

2.1.3 Challenges in Detecting TBML

Detecting TBML activities is highly challenging due to the complexity and interconnected nature of modern international trade ecosystems. Unlike conventional financial fraud, TBML transactions are often embedded within legitimate commercial operations, making suspicious activities difficult to distinguish from regular trade behavior. Criminal organizations exploit multiple jurisdictions, shell companies, intermediaries, and layered transactional structures to conceal the movement of illicit funds within normal trade systems.

Traditional AML systems primarily depend on rule-based monitoring approaches and isolated transactional analysis techniques. Although these systems are capable of identifying predefined suspicious patterns, they often fail to capture hidden relational dependencies among entities involved in large-scale trade networks. The increasing volume of cross-border financial transactions and evolving laundering strategies further reduce the effectiveness of conventional fraud detection mechanisms.

Another major challenge involves institutional privacy and regulatory compliance constraints associated with financial data sharing. Many organizations are unable to directly exchange sensitive transactional information due to confidentiality policies and legal restrictions. Consequently, modern TBML detection systems require intelligent, scalable, and privacy-preserving analytical frameworks capable of modeling complex financial relationships while supporting real-time fraud detection within large trade ecosystems.

2.2 Algorithms Used

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Federated Learning

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 User Interface

Modern financial monitoring platforms increasingly rely on scalable web application technologies to support real-time transaction analysis, fraud monitoring, and interactive visualization of financial activities. The rapid growth of digital banking and online financial services has significantly increased the demand for responsive, secure, and data-driven monitoring systems capable of handling large volumes of transactional information efficiently. Consequently, modern Anti-Money Laundering systems employ component-based web architectures and client-server communication models to provide efficient user interaction and real-time analytical capabilities.

Advanced frontend frameworks and backend service architectures play a critical role in enabling modular application development, dynamic rendering, secure authentication, and seamless integration with fraud detection engines. Such technologies support the visualization of suspicious transactional activities, alert generation, role-based access control, and automated compliance workflows within financial institutions. The integration of scalable user interface technologies with intelligent analytical systems further enhances operational efficiency and supports real-time financial crime investigation processes.

2.4.1 Modern Web Application Frameworks

Modern web application frameworks provide the architectural foundation required for developing scalable, responsive, and interactive software systems. Traditional web applications primarily relied on static page rendering and tightly coupled server-side architectures, which limited dynamic user interaction and real-time data processing capabilities. In contrast, modern frameworks adopt component-based and asynchronous rendering approaches that improve application scalability, maintainability, and user experience within large-scale distributed systems.

Frontend frameworks such as React.js enable the development of reusable and modular user interface components capable of dynamically updating application states without requiring complete page reloads. Such frameworks employ virtual Document Object Model (DOM) mechanisms and state-driven rendering principles to efficiently manage user interactions and real-time data visualization. This architectural approach significantly improves responsiveness and enables scalable dashboard development for data-intensive applications such as financial monitoring systems.

Server-side rendering and hybrid rendering frameworks such as Next.js further enhance modern web application performance by combining client-side interactivity with optimized server-side content delivery. These frameworks support efficient routing, dynamic data fetching, and improved application scalability while reducing rendering overhead for large-scale systems. Consequently, modern web application frameworks have become an essential foundation for building secure, scalable, and real-time financial monitoring platforms.

2.4.2 Component Lifecycle and Reactivity

Modern web application frameworks employ reactive rendering mechanisms to dynamically synchronize the user interface with application data and user interactions. In component-based architectures such as React and Next.js, the user interface continuously responds to changes in application state, backend responses, and client-side events. This reactive behavior enables efficient rendering of dynamic content while improving scalability, responsiveness, and maintainability within large-scale financial monitoring systems.

The component lifecycle in modern React and Next.js applications generally follows a sequence of rendering, interaction, updating, and cleanup stages. These stages collectively manage how components are initialized, rendered, updated, and removed during application execution. The overall lifecycle flow can be summarized as follows:

- **Server-Side Rendering:** Initially, application components are rendered within the server environment where data fetching and content generation are performed before transmitting the rendered output to the client browser. This approach improves initial loading performance and reduces unnecessary client-side computational overhead.

- **Hydration and Client Initialization:** After the rendered HTML content reaches the client browser, React hydrates the interface by attaching event listeners and enabling interactive functionalities within client-side components. During this stage, local states, dynamic properties, and asynchronous operations are initialized for user interaction.

- **Reactive State Updates:** As users interact with the application, component states and properties continuously change according to system events and backend responses. React employs Virtual DOM reconciliation mechanisms to compare interface changes and selectively update only the affected components instead of re-rendering the complete application interface.
- **Re-rendering and Synchronization:** Whenever state transitions occur, the framework dynamically re-renders the modified components and synchronizes the updated data with the user interface. This reactive rendering process enables real-time visualization of financial transactions, fraud alerts, and monitoring analytics within Anti-Money Laundering dashboards.
- **Unmounting and Resource Cleanup:** When components are removed from the active interface, allocated resources, subscriptions, and temporary states are released to maintain application stability and efficient memory utilization.

Such lifecycle management principles are essential in modern financial monitoring platforms where continuous transactional updates, alert generation, and real-time analytical visualization must be processed efficiently while ensuring scalable and responsive user interaction.

2.4.3 Data Visualization and Monitoring Dashboards

Understanding the theoretical aspects of data visualization is important for analyzing large volumes of financial transactions and interpreting suspicious transactional behavior within real-time monitoring systems. Visualization of analytical outputs enables financial institutions and compliance analysts to identify fraud patterns, transaction anomalies, risk distributions, and suspicious entity relationships more efficiently when compared with conventional tabular analysis methods. Interactive monitoring dashboards further improve situational awareness by enabling continuous observation of fraud alerts, transaction flows, and compliance activities without interrupting real-time analytical processes.

This work conceptualizes financial monitoring and fraud visualization through a web-based dashboard architecture developed using modern frontend technologies and analytical visualization frameworks:

- **Nextjs** is employed as the frontend framework for developing responsive and modular user interface components capable of supporting scalable financial monitoring applications.
- **Recharts** is utilized for visualizing transactional analytics, fraud classifications, risk score distributions, and temporal financial trends through interactive line graphs, bar charts, pie charts, and statistical dashboards.
- **Role-based dashboard interfaces** enable controlled access to monitoring functionalities, fraud investigation workflows, Suspicious Transaction Report (STR) generation, and real-time alert visualization according to user privileges and institutional responsibilities.
- **Interactive analytical panels** support continuous monitoring of suspicious activities, transaction histories, and fraud prediction results generated by intelligent detection systems operating within the platform.

Modern visualization systems additionally support notification mechanisms, analytical filtering, and export functionalities that assist investigators and compliance officers in conducting financial analysis and generating compliance reports for further examination. The detailed implementation and integration of these visualization technologies within the proposed TBML detection framework are discussed further in the system implementation chapter.

Summary

In this chapter, the theoretical foundations required for understanding the proposed Trade-Based Money Laundering detection framework were presented. The chapter began with an introduction to Trade-Based Money Laundering mechanisms and common laundering techniques, establishing the financial and transactional context in which illicit fund movement occurs within international trade ecosystems.

The chapter further explored the challenges associated with detecting suspicious trade activities using conventional Anti-Money Laundering approaches. Concepts related to graph-based financial modeling, heterogeneous graph representation, Graph Neural Networks, Graph Attention mechanisms, and federated learning were discussed to provide the analytical and computational foundation required for intelligent and privacy-preserving fraud detection systems.

Finally, the chapter introduced the concepts associated with modern financial monitoring interfaces, reactive web architectures, and real-time data visualization systems used for fraud analytics and compliance monitoring. The theoretical concepts discussed throughout this chapter provide the foundation for understanding the system architecture, implementation methodologies, and analytical workflows presented in the subsequent chapters.

Chapter 3

Software Requirement Specification

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

The development of an intelligent Trade-Based Money Laundering detection framework requires careful identification of the operational, computational, and functional requirements associated with the system. Since the proposed framework integrates graph-based fraud detection, federated learning, real-time monitoring, and compliance management mechanisms, defining the system requirements becomes essential for ensuring scalability, security, and efficient analytical performance. This chapter presents the Software Requirement Specification (SRS) of the proposed TBML detection framework, including the overall system description, product functionalities, user characteristics, functional and non-functional requirements, and the software and hardware resources required for implementing the proposed system.

3.1 Overall Description

The proposed system is designed as a privacy-preserving financial monitoring platform for detecting suspicious Trade-Based Money Laundering activities within large-scale transactional ecosystems. The framework combines heterogeneous graph-based fraud analysis, federated learning mechanisms, role-based monitoring dashboards, and automated Suspicious Transaction Report (STR) generation within a unified analytical environment. The system supports real-time transaction monitoring, fraud classification, and interactive visualization of suspicious financial activities to assist compliance analysts and financial institutions in identifying hidden laundering patterns efficiently.

3.1.1 Product Perspective

The proposed TBML detection framework functions as an intelligent financial monitoring and fraud analysis platform capable of integrating machine learning-based anomaly detection with interactive compliance monitoring systems. The architecture combines heterogeneous graph representation, Graph Neural Network-based analytical models, federated learning mechanisms, and modern web-based visualization technologies within a unified system environment.

The system represents financial entities such as accounts, organizations, invoices, and transactions as interconnected graph structures for relational analysis. Fraud detection modules analyze these relationships to identify suspicious transactional dependencies and hidden laundering patterns that are difficult to detect using conventional rule-based Anti-Money Laundering systems. The framework additionally integrates role-based dashboards for transaction monitoring, fraud visualization, alert management, and automated compliance reporting within financial institutions.

3.1.2 Product Functions

The primary functionalities supported by the proposed framework are summarized as follows:

- **Transaction Monitoring:** The system continuously monitors financial transactions and trade-related activities to identify suspicious transactional behavior within large-scale financial ecosystems.
- **Graph-Based Financial Analysis:** Financial entities such as accounts, organizations, invoices, and transactions are represented as interconnected graph structures for relational analysis and hidden dependency detection.
- **Fraud Classification and Risk Prediction:** Graph Neural Network-based analytical models classify suspicious transactions and generate fraud risk predictions based on transactional relationships and behavioral patterns.
- **Federated Learning Support:** The framework supports collaborative model training across multiple institutional environments while preserving sensitive financial data privacy.
- **Real-Time Dashboard Visualization:** Interactive monitoring dashboards provide visualization of transaction analytics, fraud alerts, risk distributions, and suspicious activity summaries for compliance analysts.
- **Suspicious Transaction Report (STR) Generation:** The system automatically generates structured Suspicious Transaction Reports for identified fraudulent transactions to support compliance and regulatory workflows.
- **Authentication and Role-Based Access Control:** Secure authentication and controlled access mechanisms are incorporated to manage user privileges and protect sensitive financial information within the monitoring platform.

3.1.3 User Characteristics

The proposed TBML detection framework is designed to support role-based interaction for users involved in transaction monitoring, fraud investigation, and regulatory compliance activities. Since the system handles sensitive financial information and suspicious transactional records, controlled access mechanisms are incorporated to ensure secure and authorized usage of the platform.

The major user categories supported by the system are summarized as follows:

- **Bank Compliance Team:** The bank compliance team is responsible for monitoring institution-level transactions and analyzing suspicious transactional activities identified by the fraud detection system. Authorized users can review transaction details, examine analytical dashboards, verify fraud predictions, and classify suspicious activities for further investigation. In addition, the compliance team can escalate suspicious transactions as fraudulent, mark transactions for clearance, and generate Suspicious Transaction Reports (STRs) for confirmed fraud cases.

- **Regulatory Administrator:** The regulatory administrator functions as a higher-level monitoring authority with access to suspicious transaction reports generated by participating financial institutions. The administrator can review institution-level fraud cases, analyze consolidated STR records, monitor compliance activities across multiple banks, and oversee large-scale suspicious financial activities within the monitoring ecosystem.

3.1.4 Constraints

The development and deployment of the proposed TBML detection framework are subject to multiple operational, computational, and regulatory constraints associated with financial monitoring systems. These limitations influence the scalability, analytical performance, and real-time operational capabilities of the framework.

The major constraints associated with the proposed system are summarized as follows:

- **Data Privacy and Regulatory Constraints:** Financial institutions operate under strict privacy regulations and compliance policies that restrict direct sharing of sensitive transactional information across organizations.

- **Large-Scale Transactional Data Processing:** The system must process large volumes of interconnected financial transactions and graph relationships, which may increase computational complexity and memory utilization during real-time analysis.

- **Availability of Real-World TBML Datasets:** Access to publicly available Trade-Based Money Laundering datasets is limited due to confidentiality and regulatory restrictions, making large-scale real-world experimentation challenging.

- **Dynamic Fraud Patterns:** Money laundering strategies continuously evolve over time, requiring fraud detection models to adapt to changing transactional behaviors and emerging laundering techniques.

- **Infrastructure and Computational Limitations:** Training graph-based machine learning models and processing heterogeneous financial networks require significant computational resources, especially for large-scale transactional ecosystems.

3.2 Specific Requirements

The specific requirements of the proposed TBML detection framework define the operational functionalities, performance expectations, software dependencies, and computational resources required for efficient execution of the system. These requirements ensure that the framework can support real-time fraud analysis, scalable transaction monitoring, secure user interaction, and intelligent compliance management within financial institutions.

3.2.1 Functional Requirements

The functional requirements define the major operations and services that must be supported by the proposed TBML detection framework. The primary functional requirements of the system are summarized as follows:

- The system shall authenticate authorized users before providing access to monitoring dashboards and analytical services.

- The system shall support role-based access control for bank compliance teams and regulatory administrators.

- The system shall process transactional datasets and construct heterogeneous graph structures representing financial entities and transactional relationships.
- The system shall classify suspicious financial transactions using graph-based analytical models and generate fraud risk predictions.
- The system shall support real-time visualization of transaction analytics, fraud alerts, and suspicious activity summaries through interactive dashboards.
- The system shall allow bank compliance teams to review suspicious transactions and classify them for fraud escalation or clearance.
- The system shall automatically generate Suspicious Transaction Reports (STRs) for identified fraudulent transactions.
- The system shall enable regulatory administrators to monitor fraud reports and suspicious activities generated across participating financial institutions.
- The system shall support analytical filtering and transaction search functionalities for efficient fraud investigation.
- The system shall maintain secure storage and controlled access to sensitive transactional and compliance-related information.

3.2.2 Non-Functional Requirements

The non-functional requirements define the quality attributes, operational constraints, and performance expectations associated with the proposed TBML detection framework. These requirements ensure that the system remains scalable, secure, reliable, and responsive while processing large-scale financial transactions and analytical workloads.

The primary non-functional requirements associated with the proposed system are summarized as follows:

- **Scalability:** The framework should support scalable processing of large transactional datasets and heterogeneous graph structures without significant degradation in analytical performance.

- **Real-Time Processing:** The system should support low-latency transaction analysis and fraud prediction to enable continuous real-time monitoring of suspicious financial activities.
- **Security and Privacy:** The framework should ensure secure access control, protected communication, and privacy-preserving handling of sensitive financial information and compliance-related data.
- **Reliability:** The system should maintain stable analytical performance and continuous monitoring capabilities during prolonged operational workloads and concurrent user access.
- **Availability:** The monitoring platform should provide consistent access to dashboards, fraud analytics, and reporting services for authorized institutional users.
- **Maintainability:** The system architecture should support modular development, efficient debugging, and simplified integration of additional analytical modules and monitoring functionalities.
- **Usability:** The dashboard interfaces and visualization systems should provide intuitive interaction mechanisms for compliance analysts and regulatory administrators during fraud investigation activities.

3.2.3 Software Requirements

The implementation of the proposed TBML detection framework requires multiple software technologies, machine learning libraries, backend services, database systems, and visualization frameworks to support graph-based fraud detection, federated learning, real-time transaction monitoring, and compliance management functionalities. The proposed framework integrates both analytical and monitoring components that require coordinated interaction between frontend dashboards, backend processing services, database systems, and machine learning environments. These technologies collectively support scalable fraud analysis, secure transaction management, and efficient visualization of suspicious financial activities. The software requirements associated with the proposed system are summarized in Table **??**.

Table 3.1: Software Requirements of the Proposed TBML Detection Framework

Category	Technologies / Tools	Purpose in the System
Operating System	Windows 11	Provides the development and deployment environment for backend services, databases, and analytical modules.
Programming Languages	Python, JavaScript	Used for backend development, machine learning workflows, API services, and frontend application development.
Frontend Frameworks	Next.js, React.js, Tailwind CSS	Supports responsive user interface development, dashboard visualization, and real-time monitoring functionalities.
Backend Framework	FastAPI	Facilitates asynchronous API communication between frontend dashboards and backend fraud detection services.
Database Systems	PostgreSQL, Memgraph	Stores transactional records, graph relationships, compliance information, and analytical data required for fraud analysis.
Machine Learning Frameworks	PyTorch, PyTorch Geometric	Supports deep learning operations, Graph Neural Network implementation, and graph-driven fraud analysis.
Federated Learning Framework	Flower	Enables distributed federated learning and privacy-preserving model aggregation across institutional environments.
Streaming Platform	Apache Kafka	Supports real-time transaction streaming and event-driven processing within the monitoring pipeline.
Containerization Platform	Docker	Provides containerized deployment and scalable execution environments for system integration.
LLM Integration Service	Groq LLM API	Supports automated generation of Suspicious Transaction Reports and compliance-oriented analytical summaries.

3.2.4 Hardware Requirements

The implementation of the proposed TBML detection framework requires the following hardware components to support graph processing, machine learning model execution, federated learning coordination, real-time transaction monitoring, and analytical dashboard visualization. The hardware requirements associated with the proposed system are summarized in Table **??**.

Table 3.2: Hardware Requirements of the Proposed TBML Detection Framework

Hardware Component	**Minimum Requirement**	**Purpose in the System**
Processor	Intel Core i7 / AMD Ryzen 7	Supports efficient execution of graph analytical models, backend services, and machine learning computations.
RAM	16 GB	Supports graph datasets, real-time analytical workloads, and concurrent monitoring operations.
GPU	NVIDIA RTX 3050	Accelerates Graph Neural Network training and deep learning computations.
Storage	512 GB SSD	Stores transactional datasets, graph databases, analytical models, and generated compliance reports.

Summary

In this chapter, the Software Requirement Specification of the proposed Trade-Based Money Laundering detection framework was presented. The operational functionalities, user roles, functional and non-functional requirements associated with the proposed fraud detection platform were discussed to establish the monitoring and analytical capabilities of the system.

The chapter also identified the software technologies, database systems, machine learning frameworks, and hardware resources required for implementing the proposed framework. These specifications collectively establish the operational and computational foundation required for developing the proposed TBML detection system, which is further discussed in the subsequent system design and implementation chapter.

Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter).But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects

2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter).But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
    - Table 1.1 Results of the experiment . . . .(Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table **??** is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	**ISO ALPHA 2 Code**	**ISO ALPHA 3 Code**	**ISO numeric Code**
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. **??**, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \tag{5.1}$$

The mass-energy equivalence is described by the famous equation in Eqn. **??**

$$E = mc^2 \tag{5.2}$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes

Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use tcblisting for creating the code snippets. The following example illustrates how one can customize the tcblisting to achieve the tcl script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```

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